

Software Engineering Concepts

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Topics

- 1) What is a software ?
- 2) What is Software Engineering ?
- 3) DevOps
- 4) Software Configuration Management (Git Version Control)

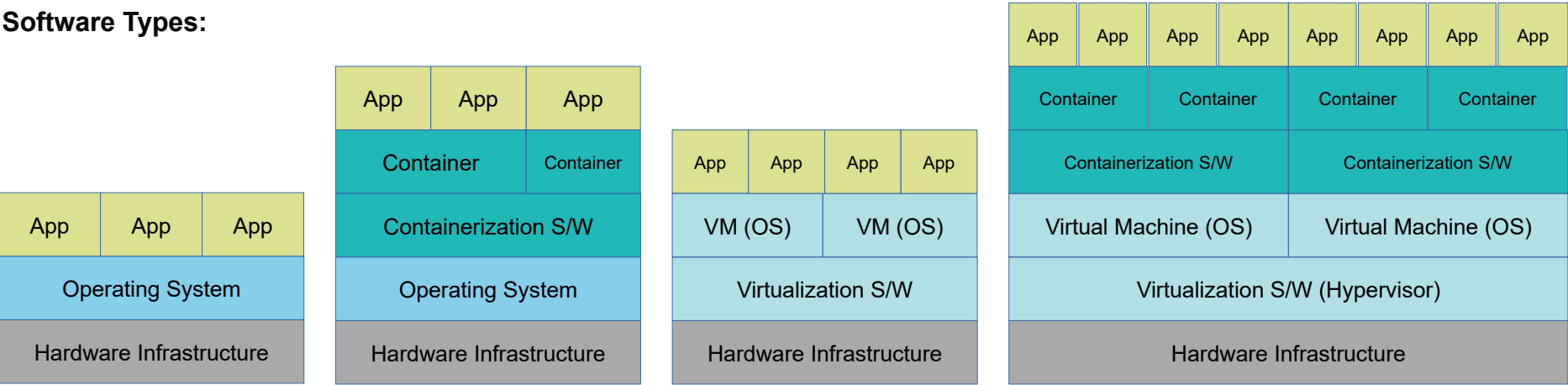
What is a software ?

collection of instructions, data, and programs used to operate a computer and execute specific tasks (IEEE)

Example: Operating Systems, Application Programs, System Services

Hardware Dependency: All software need hardware to run. Some software directly interact with the hardware e.g. System software (Operating System), Virtualization s/w, etc. Some indirectly use it (through system software) e.g. Containerization s/w, Application s/w.

Software Types:



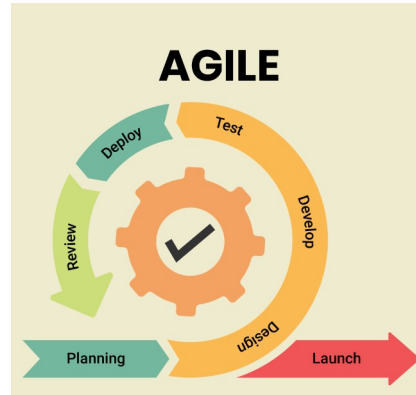
What is a Software Engineering?

collection of principles, practices and methods to develop, maintain and evolve software.



SCM ?

Over the entire development lifecycle, it is required to trace back and control changes to the software. SCM helps managing.

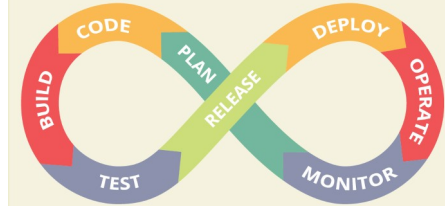


Agile ?

An iterative development lifecycle

- In traditional development, entire development was done in a single cycle
- In Agile, entire project is divided into smaller requirements, and developed in iteration.
- All development stages covered in a single iteration, Release

DEVOPS



DevOps ?

A collaboration of development and operations activities:

- In traditional development, separate team performed the development and operations.
- In DevOps, same team does both the activities

Development Tasks: Estimate, Plan, Develop

Operations Team: Deploy, Test, Release

SCM (Git Version Control)

Version Control: tracking changes to files over time.

Repository: a storage or directory holding software files.

Commit: changes to a repository files at a particular time.

Branch: a snapshot copy of a repository taken from a particular commit. Can have further changes and commits after that.

Remote Repository: centralised repository on the server.

Local Repository: A copy of remote repository kept locally on developer machine. Can be in sync or out of sync with the remote repository.

Fetch: operation that lists the remote repository changes comparing with local repository

Pull: operation that downloads and updates the remote repository changes into local repository

Merge: operation that can combine changes (series of commits) from two branches into a single branch.

